16.10.2026

Exercise 5



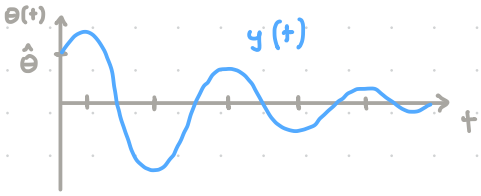
Goal Today: Transfer Functions

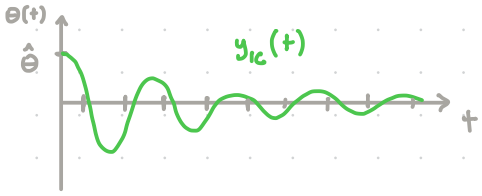
Last week:

$$y(t) = \underbrace{C \cdot e^{At} \cdot x_0}_{y_{IC}} + \underbrace{C \cdot \int_0^t e^{A(t-\tau)} B u(\tau)\, d\tau + D u(\tau)}_{y_F}$$

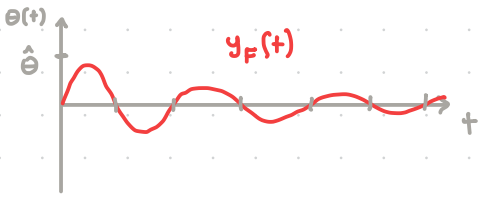
$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Full time response



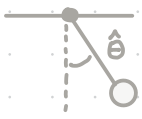


+



Initial Condition Response

- What if we just let the system evolve by itself from the initial condition:



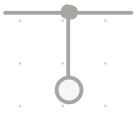
pendulum in initial condition $x_0 = \begin{pmatrix} \hat{\theta} \\ 0 \end{pmatrix}$



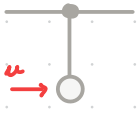
pendulum swings because it wants to go back to $x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Forced Response

- What if we start from initial condition $x_0 = 0$ and only applied our input:



pendulum with no initial condition $x_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$



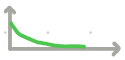
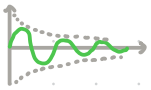
we apply input u at $t=0$

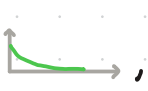
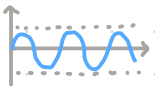
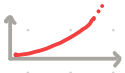


pendulum swings because of our input

Made with Goodnotes

$$y(t) = \underbrace{C \cdot e^{At} \cdot x_0}_{y_{IC}} + \underbrace{C \cdot \int_0^t e^{A(t-\tau)} B u(\tau)\, d\tau + D u(\tau)}_{y_F}$$

- Every eigenvalue λ of A adds a piece $e^{\lambda t}$ to y_{IC}:
  - → real λ: decay or growth 
  - → complex pair $\sigma \pm j\omega$: an oscillation with envelope $e^{\sigma t}$ 

- Thus, stability can also be read from A:
  - → all $\mathrm{Re}(\lambda_i) < 0$: asymptotically stable 
  - → all $\mathrm{Re}(\lambda_i) \leq 0$ and no Jordan Block: Lyapunov stable 
  - → otherwise: unstable 

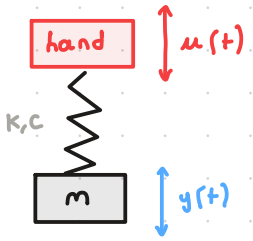
What we still can't do

$y_{IC} = C \cdot e^{At} \cdot x_0$ can now be read off of eigenvalues. But y_F is different. For every new input $u(t)$ we have to evaluate a massive integral. And even then the results don't tell us *why* the output looks the way it does...

Today: make the forced response y_F as easy as to read as the initial condition response y_{IC}. The tool for that is called a transfer function.

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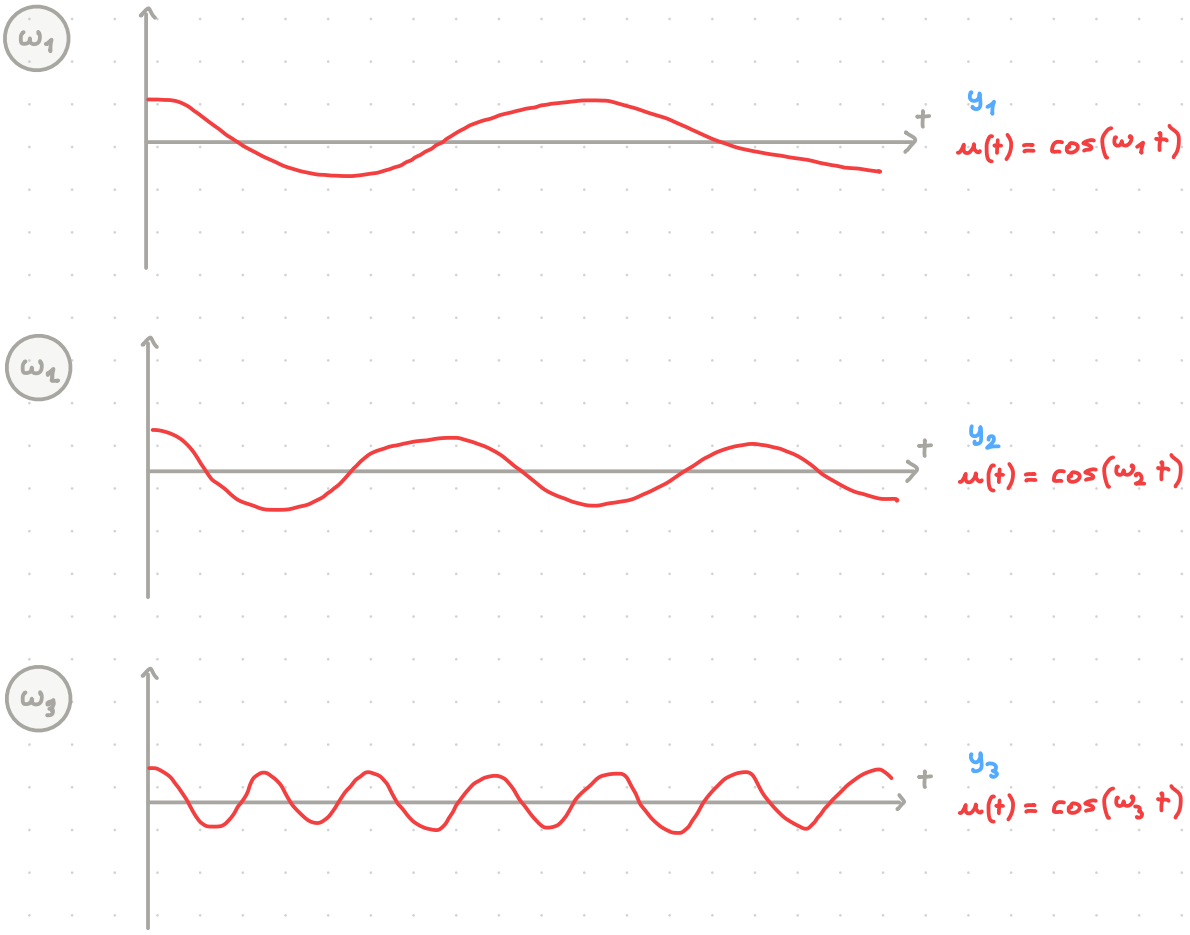
① Let's start with an experiment:



- Hold a mass on a rubber band and move your hand up and down: $u(t) = \cos(\omega t)$
  The mass position is $y(t)$. Force balance gives:

$$m\ddot{y} = K(u - y) - c\dot{y}$$

With $m = K = 1$ and $c = 0{,}4$ we get: $\ddot{y} + 0{,}4\dot{y} + y = u$
We will test 3 oscillation speeds: $\omega_1, \omega_2, \omega_3$
$\omega_1 < \omega_2 < \omega_3$



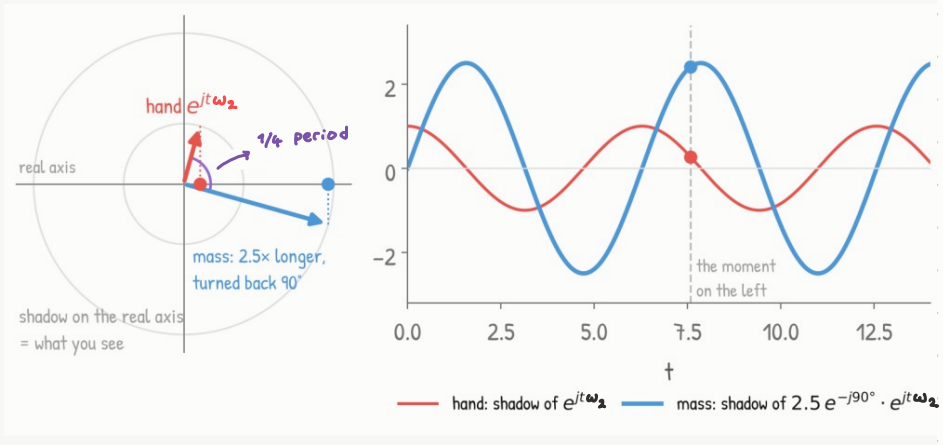
Notice: once the start-up (y_{IC}) has died out, in all three cases the mass moves as a sinusoid with exactly the frequency of the hand. Only two things change: how big the swing is ($\approx 1\times$, $\approx 2{,}5\times$, $\approx 0{,}2\times$) and how delayed it is in comparison to $u(t)$ ($\varphi_1 = 0$, $\varphi_2 = 1/4$ period, $\varphi_3 = 1/2$ period)

- Nothing in $\ddot{y} + 0{,}4\dot{y} + y = u$ says that the output has to be a sinusoid of the same frequency. So why is it? And can we predict the amplification and the delay for every ω without solving the ODE every time?

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② Stretch and delay: one complex number

Let's take the middle case ω_2. Once the start-up is gone, the system does exactly two things to the input: it **stretches** it by 2.5 and **delays** it by a quarter period. A real number can only stretch a signal, but a complex number can also delay.



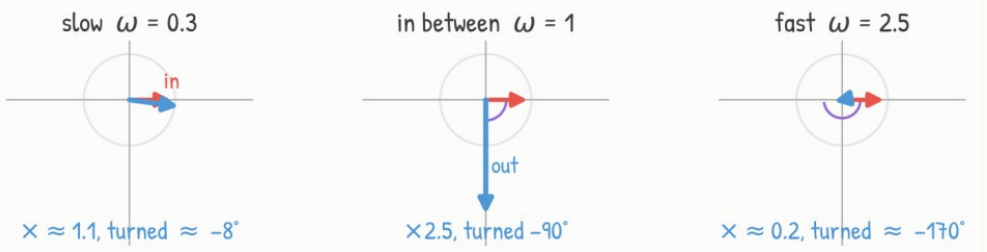
Notice: At every moment the blue arrow is the red arrow, 2.5× larger and turned back by $T/4 = \pi/2\omega_2$. That is just **one multiplication**.
Blue = $(2.5 \cdot e^{-j\pi/2\omega}) \cdot$ Red.

our system ↖

$$\sum e^{j\omega t} = (\text{complex number}) \cdot e^{j\omega t}$$

→ This is an **eigenvalue equation!!!!**
→ The signal doesn't change shape, only **scaled**.

○ All three experiments had a corresponding eigenvalue, dependent on the ω.

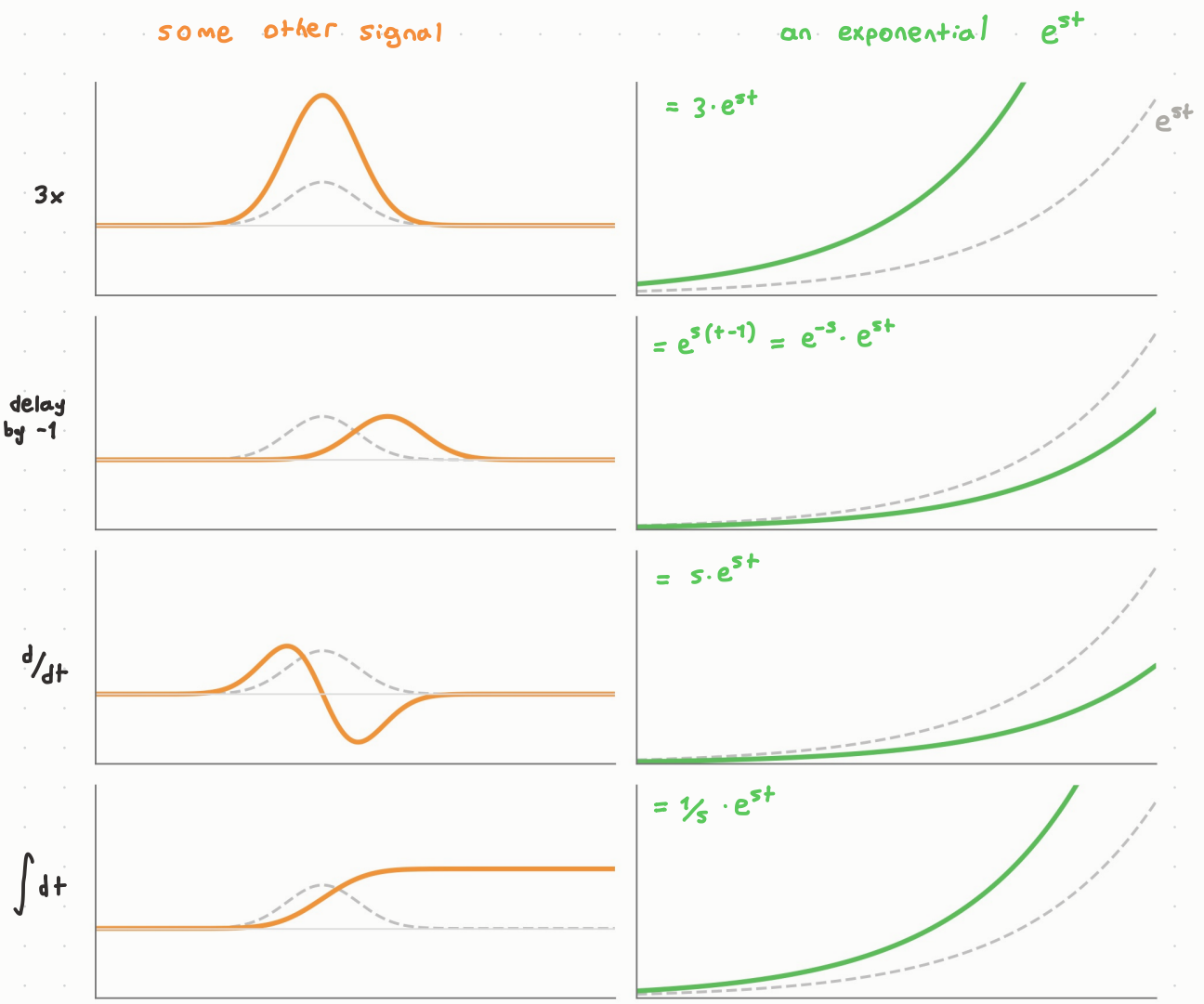


→ Is this property special to the spring, or true for every LTI system?

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③ Which inputs pass through any system without changing shape?

The spring handed $e^{j\omega t}$ as a multiple of itself. To see why, look at what an LTI system actually does to signals. When looking at system properties (week 3), we saw an LTI system could multiply by a constant, add, integrate, derivate or delay. Let's apply these to two systems:



Notice: on the right, every operation hands back the same e^{st}, only multiplied by another number.

- So feed e^{st} into any LTI system, you get the steady-state output:

$$\sum e^{st} = G(s) \cdot e^{st}$$

- The number $G(s)$ depends on the system and is a function of s. It is the transfer function

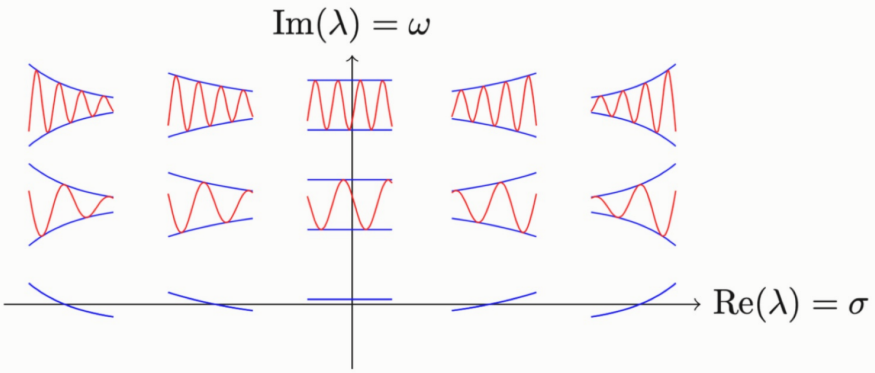
- So the spring's $e^{j\omega t}$ was just the case $s = j\omega$. The eigenfunctions of every LTI system are all e^{st}, with $G(s)$ being their eigenvalues. The three complex numbers from the experiments were: $G(\omega_1 \cdot j)$, $G(\omega_2 \cdot j)$, $G(\omega_3 \cdot j)$

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④ What does e^{st} look like?

- s is generally a complex number, $s = \sigma + j\omega$:

$$e^{st} = e^{\sigma \cdot t} \cdot (\cos\omega t + j\sin\omega t)$$



- A single complex exponential isn't a physical input. We need s and $\bar{s}$ for a real input:

$$\cos(\omega t) = 1/2 \cdot e^{j\omega t} + 1/2 \cdot e^{-j\omega t}$$

⑥ A simple system: $\dot{x} = -x + u$, $y = x$

- Let $u = e^{st}$. The last page told us:

$$y(t) = x(t) = G(s) \cdot e^{st} \quad \Rightarrow \quad \dot{x}(t) = s \cdot G(s) \cdot e^{st}$$

- Plug it in:

$$s \cdot G(s) \cdot e^{st} = -G(s) \cdot e^{st} + e^{st} \quad \Rightarrow \quad G(s) = \frac{1}{s+1}$$

- The derivatives/integrals turn into multiplications. Calculus becomes algebra. This is why transfer functions make our lives easier.

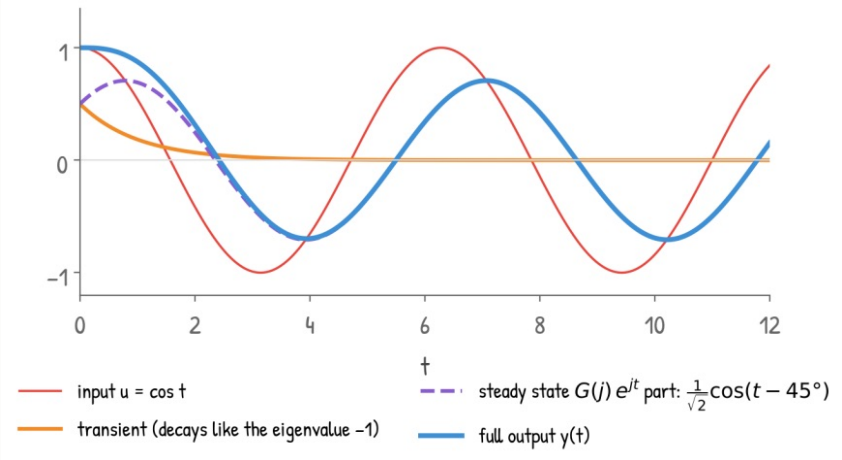
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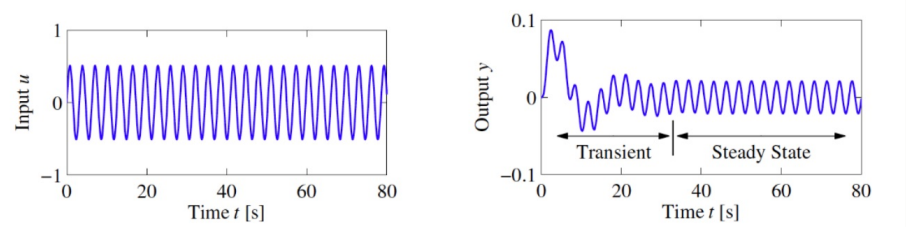
- But we should still keep in mind that the output is not just $G(s) \cdot e^{st}$. $G(s) \cdot e^{st}$ describes steady state, so when the transient has died down. The whole output is transient + $G(s) \cdot e^{st}$.

→ In this example: transient = $\left(x(0) - \frac{1}{s+1}\right) \cdot e^{-t}$

→ We don't deal with transients in CS1.

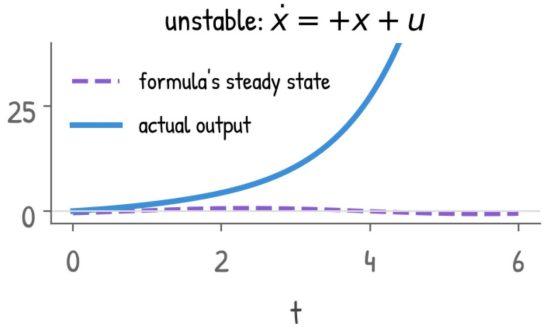
→ Transient ≠ y_{IC}



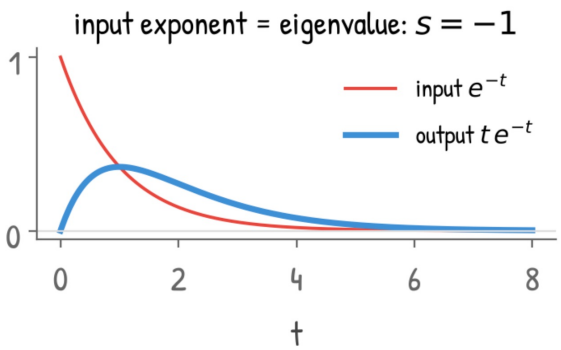


⚠ Caveats:

- In unstable systems, the transient diverges. $G(s)$ may still be computed but $G(s) \cdot e^{st}$ is never seen in the output:



- If s is equal to an eigenvalue $G(s) \to \infty$. $G(s) \cdot e^{st}$ doesn't hold. More on this next week.



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6 Any system

- After seeing the effects on a simple system, we can apply the same knowledge on a general state-space model.

$$\dot{x} = Ax + Bu \qquad y = Cx + Du \qquad ; \quad u = e^{st}$$

$$y = G(s)\cdot e^{st} = Cx + D\cdot e^{st}$$

$$x = C^{-1}(G-D)\cdot e^{st}$$

$$\dot{x} = s\cdot x = Ax + B\cdot e^{st}$$

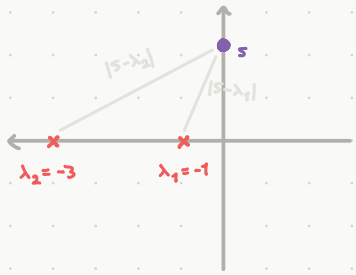
$$(s\cdot I - A)\cdot C^{-1}(G-D)\cdot e^{st} = B\cdot e^{st}$$

$$C(s\cdot I - A)^{-1}B = G - D$$

$$\Rightarrow \quad G(s) = C\cdot(sI-A)^{-1}\cdot B + D$$

Transfer Function (general form)

A picture for G(s)



- If A is diagonal with eigenvalues λ_i, everything decouples:

$$G(s) = \frac{c_1\cdot b_1}{s-\lambda_1} + \frac{c_2\cdot b_2}{s-\lambda_2} + \ldots + d$$

So if we excite the system close to an eigenvalue λ_i, that mode $\frac{c_i\cdot b_i}{s-\lambda_i}$ blows up.

You know this as resonance.

- The roots of the nominator of G(s) are called zeros.
- The roots of the denominator of G(s) are called poles.

? Your turn:

Problem: You are given a linear time-invariant system of the form:

$$\dot{x}(t) = A\cdot x(t) + B\cdot u(t),$$

$$y(t) = C\cdot x(t) + D\cdot u(t).$$

Q16 **(1 Points)**

Given:

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, C = \begin{bmatrix} 0 & 1 \end{bmatrix}, D = -1.$$

Which of the following transfer functions $G(s)$ is equivalent to the given state space system. Mark the correct answer.

- ☐ $G(s) = -\frac{s^2-3\cdot s+2}{s^2-4\cdot s+1}$
- ☐ $G(s) = -\frac{(s+2)\cdot(s-1)}{s^2-4\cdot s+1}$
- ☐ $G(s) = \frac{s^2-3\cdot s+2}{s^2-4\cdot s+1}$
- ☐ $G(s) = -\frac{s^2+3\cdot s+2}{s^2-4\cdot s+1}$

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- The whole time response to $u = e^{st}$ is:

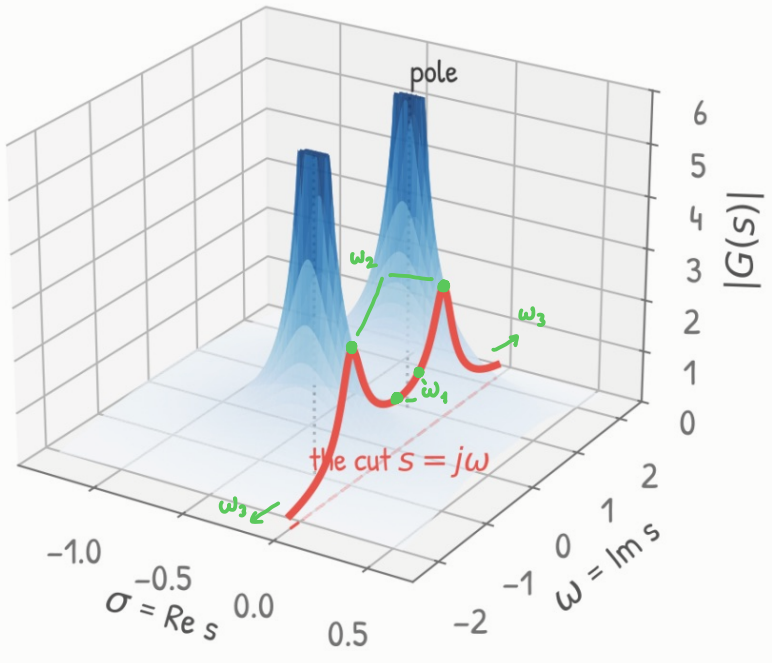
$$y(t) = \underbrace{Ce^{At}\left[x(0) - (sI-A)^{-1}B\right]}_{\text{transient} \to 0 \text{ if stable}} + \underbrace{\left[C(sI-A)^{-1}B + D\right]e^{st}}_{\text{steady-state response}}$$

(7) Back to the spring

$$\ddot{y} + 0{,}4\dot{y} + y = u \quad \Rightarrow \text{(week 2)} \quad \dot{x} = \begin{pmatrix} 0 & 1 \\ -1 & -0{,}4 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u, \quad y = \begin{pmatrix} 1 & 0 \end{pmatrix} x$$

$$G(s) = C \cdot (sI - A)^{-1} \cdot B + D \quad \Rightarrow \quad G(s) = \frac{1}{s^2 + 0{,}4s + 1}$$

- $s = 0$ (move the hand and hold it) $\Rightarrow G(0) = 1 \Rightarrow$ mass shifted by same amount ✓
- $|s|$ large (very fast hand) $\Rightarrow G(s) \to 0 \Rightarrow$ mass doesn't move ✓



$|G(s)|$ is how much our signal is amplified

Made with Goodnotes

8 Analyzing Sinusoids

- For sinusoids like $u = \cos(\omega t)$, we can simplify the steady-state output even further:

$$\cos\omega t = \tfrac{1}{2} e^{j\omega t} + \tfrac{1}{2} e^{-j\omega t} \Rightarrow y_{ss} = \tfrac{1}{2} \cdot G(j\omega) \cdot e^{j\omega t} + \tfrac{1}{2} \cdot G(-j\omega) \cdot e^{-j\omega t}$$

$$G(j\omega) = M \cdot e^{j\varphi} \Rightarrow y_{ss} = \tfrac{1}{2} M \left(e^{j(\omega t + \varphi)} + e^{-j(\omega t + \varphi)} \right) = M \cos(\omega t + \varphi)$$

$$\boxed{y_{ss} = |G(j\omega)| \cdot \cos\left(\omega t + \angle G(j\omega)\right)}$$

? Your Turn

Problem: Consider a linear time-invariant system with transfer function $G(s)$ given by

$$G(s) = \frac{3s}{s^2 + 4s + 5}.$$

Further, the signal $u(t) = \sin(t)$ is applied as the input to the system $G(s)$. It is known that the corresponding steady-state output signal is given by $y_{ss}(t) = M \sin(\omega t + \phi)$.

Q14 (1.0 Points)
What is the value of M in the above expression of $y_{ss}(t)$?

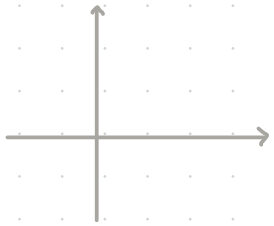
Q15 (1.0 Points)
What is the value of ϕ in the above expression of $y_{ss}(t)$?

$$|z| = \sqrt{\mathrm{Re}(z)^2 + \mathrm{Im}(z)^2}$$

$$\angle z = \arctan\left(\mathrm{Im}(z) / \mathrm{Re}(z)\right)$$

or

plot it:



Made with Goodnotes

(9) Any input is a sum of exponentials

- So far we only fed $u = e^{st}$ into our systems. But we had a reason to do so.

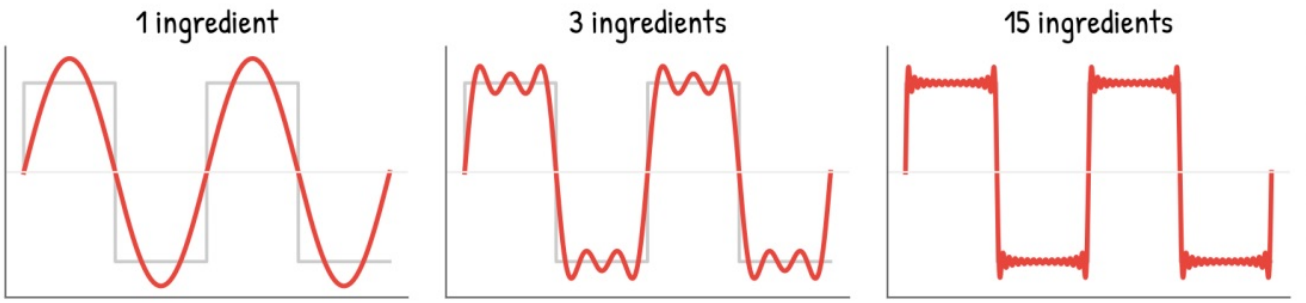
e^{st} are eigenfunctions of LTI systems

⇓ — Lin Alg

they form a complete set

⇓

we can build any function as a sum of e^{st}'s



$$u(t) = \sum_k \underbrace{U_k}_{\text{scaling factor, e.g. } 1/2 \text{ for } \cos\omega t} e^{s_k \cdot t} \quad \Rightarrow \quad y_{ss} = \sum_k \underbrace{G(s_k)}_{\text{every term has its own TF}} U_k e^{s_k \cdot t}$$

For the curious:

The operation that finds the recipe U for any signal is called the Laplace Transform. You will learn it in Ana III. Once you learn it, you can find $G(s)$ as $G(s) = \frac{Y(s)}{U(s)}$ with $Y(s) = \mathcal{L}(y(t))(s)$, $U(s) = \mathcal{L}(u(t))(s)$

Made with Goodnotes

(10) Relations between $G(s)$ and state-space

- We design control systems using $G(s)$ but computers need state-space. Thankfully, there are 2 easy formulas to transition.

→ If A is diagonal.

$$G(s) = \frac{P_1}{s-\lambda_1} + \frac{P_2}{s-\lambda_2} + \dots + \frac{P_n}{s-\lambda_n} + d \iff A = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \quad B = \begin{bmatrix} \sqrt{P_1} \\ \vdots \\ \sqrt{P_n} \end{bmatrix}$$

$$C = \begin{bmatrix} \sqrt{P_1} & \cdots & \sqrt{P_n} \end{bmatrix} \quad D = d$$

→ If A is in the controllable canonical form:

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d \iff A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & 1 \\ -a_0 & -a_1 & & \cdots & a_{n-1} \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

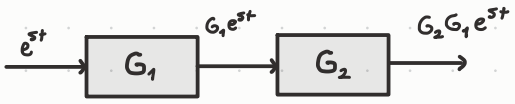
$$C = \begin{bmatrix} b_0 & b_1 & \cdots & b_{n-1} \end{bmatrix} \quad D = d$$

(11) Interconnections

- Interconnections of TF's are also easy to calculate.

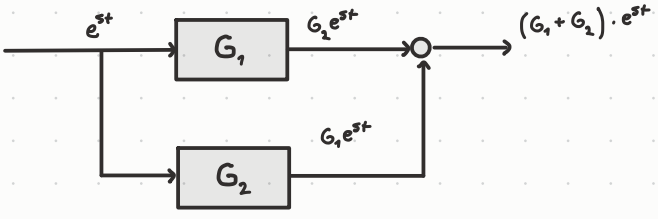
→ Serial

$$G(s) = G_2(s) \cdot G_1(s)$$



→ Parallel

$$G(s) = G_2(s) + G_1(s)$$



Made with Goodnotes